

Parkbnb V3 — First-Principles Mechanism Invention

Dossier

Smart Parking Lock · Clean-sheet mechanical architecture · August 2026

Brief: invent a new mechanical architecture for a smart parking lock. Motor must never carry structural load, ideally must not move the barrier at all. Minimum parts, 10-year outdoor life, Indian fabrication (laser + bend + weld), off-the-shelf components.

1. First Principles — Deleting the Assumptions

Assume parking locks were never invented. What does physics actually demand?

1.1 The energy question (the whole design turns on this)

Barrier: **m = 5 kg**, CG at **0.25 m** from pivot, tip at **0.45 m**, inertia **J ≈ 0.4 kg·m²**.

Energy to raise the barrier 0° → 90°: $E = m \cdot g \cdot h = 5 \times 9.81 \times 0.25 = 12.3 \text{ J}$

12.3 joules. One AA cell stores ~10,000 J. A car tyre rolling 30 mm delivers ~100 J. A human blink of effort is more than this.

Therefore the motor in every existing parking lock does not exist to supply energy. It exists because those designs chose a geometry in which the actuator must deliver the *full torque continuously through the stroke, and then hold position*. Energy was never the constraint. Torque delivery and state-holding were — and both can be solved with geometry instead of electricity.

1.2 The four questions, answered

Question	Answer that drives the architecture
Can gravity do the work?	Yes, one way (down). Free.
Where does the <i>up</i> energy come from?	The same 12.3 J the barrier gave up when it fell — if a spring catches it. A counterbalance is not a "helper"; it is an energy see-saw with a nearly free round trip.
Can geometry replace the motor?	Yes. Once the spring stores the fall energy, only ~10% is missing (friction + bias). That deficit is ~1.5 J.
Where does 1.5 J come from, free?	The car. It arrives and departs exactly when the state must change, and can spare 100 J without the driver noticing.
Can one part do several jobs?	The barrier can be its own treadle, its own cam, its own striker carrier.
Can structural load bypass moving parts?	Yes — with fold direction + hard stops, vehicle load never touches spring, latch or actuator.

1.3 The reframe

A parking lock is **not a lifting machine**. It is a **two-state latching machine** whose state changes are paid for by gravity and by the vehicle. The electronics' only job is to *permit* a state change — a pulse, not a stroke.

2. Concept Generation — 10 Architectures

C1 Direct Motorized Rotation (the incumbent, as baseline)

Principle: worm gearmotor rotates the barrier through its full arc; worm self-locks to hold.

Motor → Worm → Wheel → Barrier ∪ Load: Barrier → Wheel → Worm → Motor mounts

Advantages: simple control, proven, arbitrary duty. **Disadvantages:** motor+gearbox in *both* the motion and structural path; largest BOM block; gear wear; battery must run a motor; stall/jam is a service call. **Manufacturing:** medium. **Moving parts:** ~8. **Patent potential:** none (saturated art).

C2 Leadscrew / Self-Locking Actuator

Motor → Screw → Nut → Crank → Barrier Load: Barrier → Crank → Nut → Screw (locked)

Advantages: holds without power; high mechanical advantage. **Disadvantages:** motor still runs the *entire* stroke (violates the brief); screw exposed to grit; slow; back-driven impact through nut threads. **Manufacturing:** medium. **Moving parts:** ~7. **Patent:** none.

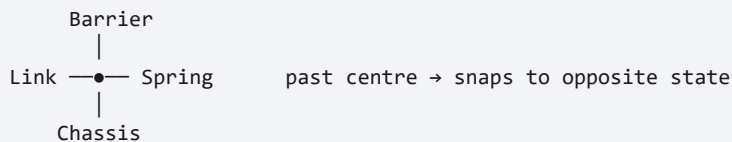
C3 Rising Bollard / Telescopic Post

Post ↑↓ inside sleeve Load: Post → Sleeve → Pit walls → Ground

Advantages: superb structural path (pure shear into a buried sleeve); very strong. **Disadvantages:** requires **excavation** — kills surface-mount retrofit, i.e. the entire market. Sleeve fills with water and silt. **Manufacturing:** hard (civil work). **Moving parts:** 3. **Patent:** none. ❌ Out of scope.

C4 Over-Center Toggle (bistable snap)

Principle: barrier is one link of a toggle; spring is stable UP and stable DOWN, unstable between. Actuator only pushes past dead-centre; the spring completes the throw.



Advantages: actuator energy $\approx$ only what crosses centre; self-holding at both ends by geometry. **Disadvantages:** snap action is violent (uncontrolled energy release — safety); a toggle used as the *security* lock is defeated by hammering it past centre; tuning drifts with wear. **Manufacturing:** easy. **Moving parts:** ~6. **Patent:** low. *Verdict: reject as a lock; the "cross-centre trip" idea survives as the release principle.*

C5 Constant-Force Spring + Cosine Cam ("weightless barrier")

Principle: gravity torque varies as $12.3 \cdot \cos \theta$. Wrap a cable on a **cam whose radius follows $\cos \theta$** , pull with a coil spring → spring torque tracks gravity *exactly*, at every angle.

Spring = cable = (CAM $r(\theta) = R \cdot \cos \theta$) — Barrier
Net torque ≈ 0 everywhere → barrier stays wherever you leave it

Advantages: true weightlessness; the cam is a **2D laser-cut plate — zero machining**; a ₹30 coil spring replaces ₹300 of torsion springs; the net torque curve becomes a *design variable* editable in CAD. **Disadvantages:** cable is a wear/fatigue item, must be captive in a groove; single-cable failure = sudden loss of balance. **Manufacturing: very easy.** **Moving parts:** 4. **Patent:** low alone (anglepoise/garage-door art) — **high in combination** (see C10).

C6 Four-Bar Rising Arc

Advantages: lower folded height; coupler curve shapeable. **Disadvantages:** 4 pivots = 4× wear, slop and cost; harder to seal; no gain a single pivot + cam cannot already give. **Manufacturing:** medium. **Moving parts:** ~8. **Patent:** low.

C7 Self-Energizing Wedge Chock

Principle: a wedge whose ground reaction increases with pushing load. **Advantages:** structurally elegant; zero holding power. **Disadvantages:** **self-energizing = release force grows with load** — the actuator can never release it after a hard push. One direction only. Needs friction that wet/oily tarmac does not guarantee. **Moving parts:** 2. **Patent:** medium. ❌ Fails releasability.

C8 Rotary Drum Barrier



dirt/water shed by rotation; slot faces down when open

Advantages: excellent ingress protection (no exposed hinge line); one part is structure + enclosure + seal. **Disadvantages:** rolled/formed shell needs tooling (breaks the no-tooling rule); large swept volume; heavy. **Manufacturing:** hard at low volume. **Moving parts:** 3. **Patent:** medium. *Revisit at 10k+ volume.*

C9 Human-Powered Counterbalance (V2-M, current baseline)

Advantages: cheapest possible; no actuator except the release solenoid; proven. **Disadvantages:** requires a human action every departure — the one thing the ₹25k incumbent does *not* require. A competitive weakness, not an engineering one. **Moving parts:** 5. **Patent:** low.



C10 TreadCycle: Vehicle-Cycled Neutral-Balance Barrier (VC-NBB)

Principle — the invention. Combine three individually unremarkable things to remove every remaining actuator:

1. **Cam-cable counterbalance (from C5)** tuned to ~88% — the barrier is nearly weightless but keeps a deliberate *fall* bias.
2. **The barrier's last 30 mm of travel is a treadle stroke.** Lying flat, it stands 30 mm proud. The next wheel presses it those 30 mm onto steel stops — and that stroke **winds a small kick spring through a one-way pawl**. The vehicle refuels the mechanism every time it passes.
3. **One solenoid releases whichever latch is engaged** (up-latch or down-latch) — only one ever is. One actuator, two functions, one app button.

U-BARRIER ——— { cam-cable counterbalance (88%, fall-biased)
(also treadle, { kick spring ← one-way pawl ← 30 mm treadle stroke (wheel)
also cam plate, {
also striker) { rotary claw latch ↔ down pawl ← ONE solenoid (pulse only)

Vehicle impact → over-travel HARD STOP → chassis → anchors → concrete
Wheel 1000 kg → bottom-out STEEL STOPS (spring/pawl/latch never see it)

Cycle: unlock pulse → barrier falls by gravity → car rolls over, pressing the treadle stroke → **spring charged, barrier held down** → car clear → release → barrier rises gently on stored energy → claw latch snaps → SECURE. Every pass pays for the next rise.

Advantages: no motor anywhere, ever; no human lift; fully automatic both directions; actuator moves ~4 mm and only unblocks a pawl; wheel load, vehicle impact and spring force are three *separate* paths that never meet; the barrier is one part doing four jobs. **Disadvantages:** the treadle stroke is a moving interface at ground level (grit/water); auto-rise needs a departure trigger; barrier must be rated for wheel load. **Manufacturing:** easy — laser, bend, weld, one shaft, catalogue springs and latch. **Moving parts:** 6. **Patent potential:** high — the

combination "*vehicle-stroke-charged counterbalanced barrier with load-isolating bottom-out stops and a single state-agnostic release actuator*" is not, to my knowledge, existing parking-lock art.

3. Concept Comparison (1 = poor, 5 = excellent)

	C1 Motor	C2 Screw	C3 Bollard	C4 Toggle	C5 Cam-CB	C6 4-Bar	C7 Wedge	C8 Drum	C9 Manual	C10 Tread
Power (less is better)	1	2	2	5	5	4	5	3	5	5
Simplicity	2	2	3	4	5	2	5	4	5	4
Reliability	3	3	4	3	4	3	2	4	4	4
Maintenance	2	2	2	4	4	3	4	4	5	4
Weight	2	3	1	4	5	3	4	2	5	4
Cost	1	2	1	4	5	3	5	2	5	4
Manufacturability (India)	3	3	1	5	5	3	5	2	5	5
Weather resistance	3	2	2	4	4	3	3	5	4	3
Safety	4	4	3	1	5	4	3	3	5	4
Serviceability	2	2	1	4	5	3	4	3	5	4
Service life	3	3	5	3	4	3	3	4	4	4
Patentability	1	1	1	2	2	2	3	3	1	5
Total (60 max)	27	29	26	43	53	36	46	39	53	50

Reading the table honestly: C5 and C9 score highest because they are the *simplest*. C10 scores 50 and is the only concept that reaches **fully automatic with zero motor** — it spends 3 points of weather-resistance and simplicity to buy the one feature the market actually compares against the ₹24,999 incumbent. C10 *contains* C5 and degrades gracefully to C9 if the treadle subsystem is deleted. **That containment is why it wins:** it is not a riskier bet, it is the same bet with an optional extra stage.

4. Force & Load Analysis (C10)

4.1 Three isolated load paths

- ① VEHICLE IMPACT (barrier up, someone noses in)
 $2000 \text{ N at } 0.45 \text{ m} = 900 \text{ N}\cdot\text{m}$
 Barrier folds TOWARD THE STREET, so impact rotates it PAST vertical into a hard stop.
 Path: Barrier → over-travel stop → chassis → 4×M10 anchors → concrete
 Seen by latch: ZERO.
- ② WHEEL DRIVE-OVER (barrier flat, 1000 kg)
 Barrier bottoms on steel stops at end of the 30 mm treadle stroke.
 Path: Tyre → barrier skin → bottom-out stops → chassis rails → anchors → concrete
 Seen by kick spring / pawl / latch: ZERO (they are past their travel limit).
- ③ TAMPER (human forcing the raised barrier down)
 $700 \text{ N two-handed at } 0.47 \text{ m} = 329 \text{ N}\cdot\text{m} \rightarrow \text{at } 100 \text{ mm striker radius} = 3.3 \text{ kN on the claw}$
 Door-class automotive latch rated 9–11 kN → SF ≈ 3.
 This is the ONLY case the latch is designed for.

4.2 Component force ledger

Member	Peak force/torque	Governing case	Margin
Pivot shaft Ø20 EN8	≤ 150 N·m bending	residual after hard stop	SF > 2.5
Over-travel hard stop	900 N·m	vehicle impact	gusseted, SF 2
Bottom-out stops	9.8 kN	wheel	direct bearing into chassis
Rotary claw + striker	3.3 kN	tamper	SF ≈ 3
Counterbalance cable	≈ 155 N	static	2 mm rope, 3 kN MBL, SF > 15
Kick spring	2.5 J, ~25 N·m	charging	≤ 40% yield
Treadle pawl	123 N	charging	trivial
Solenoid	≤ 5 N over 4 mm	pawl release	the weakest, smallest, cheapest part — <i>by design</i>

Weakest member (deliberate): the counterbalance cable. It is the one part whose failure changes behaviour — see §8 FMEA and the two-cable redundancy decision.

4.3 Governing numbers (computed)

Quantity	Value
Raise energy	12.3 J
Counterbalance setpoint	88% → net 1.47 N·m
Tip force during rise	3.3 N ≈ 0.33 kg-f (a finger stops it)
Rise time (undamped)	0.9 s → tuned to ~2 s with pivot friction
Impact energy if it strikes a hand	2.3 J — must be verified against door-safety limits (§11)
Kick-spring energy required	1.47 J deficit ÷ 60% eff = 2.45 J
Treadle: 30 mm stroke @ 200 mm radius	82 N (8.3 kg-f) = 2.3% of one 3500 N wheel — the driver cannot feel it

5. Optimization Loop — Removing Parts Until It Breaks

Pass 1 — delete the drive motor. Replaced by gravity (down) + stored energy (up). ✓

Pass 2 — delete the torsion springs. Replaced by cam + cable + 2 coil springs: cheaper (₹120 vs ₹300), exact cosine tracking, profile editable in CAD. ✓

Pass 3 — delete the rotary damper. At 88% counterbalance the descent is already gravity-limited (~1 s, 2.3 J into rubber stops). ✓

Pass 4 — delete the kick spring? ✗ Reverted. Tried setting the counterbalance *rise*-biased so the barrier self-raises with no kick spring. But then nothing makes it fall: gravity is cancelled, and a barrier folding toward the street cannot be pushed down by an approaching car. The alternative — folding toward the space so the car noses it down — puts the full 900 N·m vehicle load onto the latch, destroying load isolation. **The kick spring is therefore not decoration; it is what buys fall-bias and latch protection simultaneously.** Kept.

Pass 5 — delete the down-latch. Merged into the release/timing module: the part that holds the barrier down is the escapement pawl. One part, two functions. ✓

Pass 6 — delete the second solenoid. One solenoid, one release bar, both pawls. Only one latch is ever engaged, so a single pulse is always unambiguous. ✓

Pass 7 — delete the separate treadle plate. The barrier itself is the treadle. ✓

Pass 8 — delete the separate cam plate. The cam profile is cut into the barrier's own pivot bracket. One laser part carries the pivot bore, cam groove, striker mount and magnet pocket. ✓

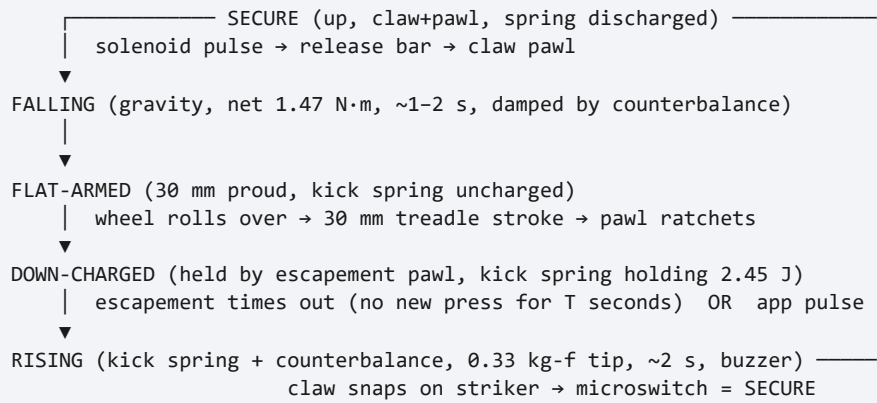
Stop condition reached: every remaining part fails a delete test.

6. Final Architecture

6.1 Exploded view (top → bottom)

- 1 U-Barrier weldment (tube arch + tread plate) ← barrier, treadle, impact member
- 2 Pivot bracket L & R (laser) ← pivot bore + cam groove + striker + magnet
- 3 Striker pin Ø10 hardened
- 4 Pivot shaft Ø20 EN8 + 2 × Oilite bushings + circlips
- 5 Counterbalance cassette: 2 × wire rope + 2 × coil springs + turnbuckle adjuster
- 6 Kick spring + one-way pawl + ratchet segment
- 7 Escapement / delay-release module (also the down-latch)
- 8 Rotary claw latch (automotive, door class) + release bar
- 9 Mini solenoid (4 mm stroke, ≤ 5 N)
- 10 Central hub ≤ 120 mm — houses 6-9 + PCB + 18650 ← also the magnet shroud
- 11 Chassis rails + over-travel hard stop + bottom-out stops (laser + bend + weld)
- 12 Base plate + 4 × M10 wedge anchors + rubber pads

6.2 State diagram



6.3 Sequences

Opening: authenticated command → 1 s warning tone → solenoid pulse (≤ 1 A, 200 ms) → claw releases → barrier falls to flat-armed.

Entry: car drives in; each axle presses the 30 mm stroke; first press charges the kick spring, later presses free-wheel on the ratchet; barrier bottoms on steel stops under load.

Closing: escapement (or app) releases the down pawl → barrier rises in ~2 s → cam ramp cocks the claw → claw snaps, pawl drops → microswitch confirms → app shows SECURE.

Exit: same as opening; the exit pass recharges the spring, so the barrier re-secures the empty bay by itself.

7. Materials

Part	Material	Reason
Barrier arch	ERW steel tube Ø32 × 2 mm	Impact energy absorption, light
Tread plate	2.5 mm HR sheet, ribbed	Wheel load, weldable
Pivot brackets / cam	5 mm laser-cut MS	Cam groove needs section, no machining
Pivot shaft	EN8 (C45)	385 MPa yield, ground stock, cheap
Bushings	Oilite bronze	Self-lubricating, grit-tolerant, replaceable
Counterbalance rope	2 mm SS 7×19 wire rope	3 kN MBL, corrosion, flexible
Springs	EN 10270-1 SH music wire, zinc + wax	Fatigue at ≤ 40% yield, outdoor
Claw & striker	Case-hardened steel (as supplied)	Wear + 3.3 kN tamper
Chassis / base	2.5–3 mm galvanised or powder-coated MS	Cost, weldability
Stops	Steel + replaceable rubber pad	Sacrificial, serviceable
Fasteners	SS304 / zinc	Coastal, monsoon

8. FMEA

#	Failure mode	Cause	Effect	Detection	Mitigation
1	Counterbalance cable break	Fatigue at cam groove	Barrier heavy; if up then unlocked, falls fast (2.3 → 12 J)	Rise time out of band; user notices weight	Two independent cables + springs (₹100) — single failure leaves 50% balance; rubber stops sized for the unbalanced case
2	Kick spring / pawl failure	Fatigue, grit	No auto-rise	Up-microswitch never confirms → alert	Degrades to C9: barrier liftable by hand (0.33 kg-f residual) — product still works
3	Escapement jams	Silt, ice	No auto-close (or premature close)	Timeout mismatch vs latch switch	Escapement is a <i>bolt-on module</i> : delete it and the app pulse closes the barrier
4	Claw jam	Grit/ice	Will not unlock	Solenoid fired, no motion on Hall → alert	Drainage slots, sealed hub, key override rotates the pawl directly
5	Solenoid burnout	Coil, over-duty	No remote unlock	No current pulse	200 ms pulse duty; key override
6	Treadle stroke silts up	Sand, leaves	Spring never charges	Same as #2	Open-bottom design (debris falls through), 8 mm clearances, no blind pockets
7	Bushing wear	Years, grit	Slop, striker misalign	Service inspection	±1.5 mm latch capture window; bushings press-out replaceable
8	Hard-stop deformation	Repeated strikes	Lock geometry shift	Visual + impact log	Gusseted stop, replaceable rubber pad
9	Barrier rises against an obstruction	Pet, foot, bumper	Contact	—	Inherent: 0.33 kg-f tip force , 2 s rise, buzzer. Not a "safety system" — a <i>safety property</i> of low stored energy
10	Water in electronics	Seal failure	Dead unit	Battery telemetry	Hub above flood plane, IP55 potted PCB, drained

9. DFM Recommendations

- **Three laser thicknesses only:** 2.5 mm (skin/chassis), 5 mm (brackets/cam), 3 mm (base).
- **No CNC milling anywhere.** The cam is a laser profile; the only turned parts are the shaft and striker pin (centreless-ground bar, cut to length).
- **Tab-and-slot self-fixturing** on every weld joint — removes fixture cost and holds the 0.3 mm pivot-bore alignment.
- **One shaft size, one bushing size, one fastener family** (M8 structural, M6 covers).
- **Buy, don't make:** latch, springs, rope, bushings, anchors, solenoid, escapement/damper. Make only 11 laser parts and 2 turned parts.
- Powder coat after weld; mask bores; press bushings last.

Indicative BOM: steel ₹750 · latch ₹300 · counterbalance cassette ₹150 · kick spring + pawl ₹90 · escapement module ₹200 (optional) · solenoid ₹120 · key override ₹100 · PCB ₹500 · battery ₹250 · hardware ₹200 → **₹2,460 without escapement, ₹2,660 with.** Holds the ₹2–3k target *and* deletes the motor.

10. Assembly, Maintenance, Prototype & Test

Assembly (target 22 min): weld chassis → weld barrier → powder coat → press bushings → shaft + circlips → hang barrier → fit cam ropes, tension to spec with turnbuckle → fit kick spring & pawl → fit latch, set striker shim to ±1.5 mm → fit hub, PCB, battery → function test 20 cycles → anchor template + packaging.

Maintenance: grease bushings and inspect ropes at 12 months; rubber pads at ~5 years; battery at ~3 years. All from the top after removing the hub cover — **no need to unbolt the unit from the ground.**

Stage	Build	Question it answers
P0	Cardboard + plywood 1:1 linkage	Fold direction, hub height, straddle clearance
P1	Steel mule, counterbalance only, no electronics	Does the cam profile hold ≤ 0.5 kg-f across 90°?
P2	Add kick spring + treadle	Does one wheel pass reliably store 2.45 J?
P3	Add latch + solenoid + escapement	Full autonomous cycle, 500 cycles
P4	Pilot 10 units, real driveways	Grit, monsoon, user behaviour

Test	Requirement
Cycle life	20,000 cycles, force drift < 15%
Drive-over	1,000 kg per wheel × 5,000 passes, zero permanent set
Impact	2,000 N at 450 mm — hard stop deforms, latch must not release
Tamper	700 N two-handed jerk × 50, pry bar, hammer
Magnet attack	N52 magnet on all six faces — must not release
Ingress	IP55 spray + dust; 500 h salt spray
Thermal	−5 °C to +60 °C, verify counterbalance and escapement timing
Safety	Rise force ≤ 0.5 kg-f, obstruction stall test, third-party check vs powered-door impact limits
Battery	≥ 3 years modelled from measured sleep current

11. Honest Risk Register

1. **The treadle stroke is the riskiest new interface** — a ground-level moving joint in a grit environment. Mitigated by open-bottom geometry (debris falls out, nothing to clog) and by the fact that its failure degrades the product to the working C9 baseline rather than bricking it. **This must be the first thing P2 tries to destroy.**
2. **Auto-rise safety needs certification, not assertion.** 2.3 J and 0.33 kg-f are low, but a barrier that rises unattended in a public driveway deserves an independent check against powered-gate/door impact standards before pilot sale.
3. **Patent claims must be searched before filing.** Cam counterbalance, ratchets and rotary latches are all old art individually. What may be defensible is the *combination*: vehicle-stroke charging + load-isolating bottom-out stops + a single state-agnostic release actuator on a counterbalanced parking barrier. Treat §2-C10 as the disclosure draft for a prior-art search, not as a granted claim.
4. **Escapement timing in cold/silt** is the least predictable part; it is deliberately a bolt-on module so the product can ship without it.

12. Future Improvements

Solar lid on the hub (already the only sun-facing surface) · LTE/LoRa for the commercial SKU · load-cell in the tread plate to sense occupancy for free · drum-shell variant (C8) once volume justifies tooling · fleet analytics from the impact log · bike/two-wheeler variant using the same hub and a shorter arch.